Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Double G Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
6555 Blue Gum AVE	Modesto	Stanislaus	95358

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1916

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X012-X028-X018-XXXX X012-X028-X021-XXXX X012-X028-X040-XXXX X012-X028-X041-XXXX

B. OPERATORS

Gwerder, Frank

Operator name: Gwerder, Frank

Telephone no.: Landline: (209) 579-5723 Cellular:

Mailing Address Number and Street	City	State	Zip Code
825 S Hart RD	Modesto	CA	95358

This operator is responsible for paying permit fees.

C. OWNERS

Genasci, Bill

Legal owner name: Genasci, Bill

Telephone no.: Landline: (209) 522-7859 Cellular:

Mailing Address Number and Street	City	State	Zip Code
6555 Blue Gum AVE	Modesto	CA	95358

Genasci, Jim

Legal owner name: Genasci, Jim

Telephone no.: Landline: (209) 522-7859 Cellular:

Mailing Address Number and Street	City	State	Zip Code
6555 Blue Gum AVE	Modesto	CA	95358

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	675	225	0	0
Number under roof	1,000	100	0	0	0	0
Maximum number	1,000	100	675	225	0	0
Average number	1,000	100	675	225	0	0
Avg live weight (lbs)	1,400	1,450	1,000	700		

Predominant milk cow breed: Holstein

Average milk production: 74 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 36,771.33 *tons per reporting period*

Total nitrogen from manure: 445,177.94 *lbs per reporting period*

After ammonia losses (30% loss applied): 311,624.56 *lbs per reporting period*

Total phosphorus from manure: 74,769.41 *lbs per reporting period*

Total potassium from manure: 194,137.23 *lbs per reporting period*

Total salt from manure: 493,845.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 15,609,000 *gallons*

Total nitrogen generated: 79,334.79 *lbs*

Total phosphorus generated: 8,819.07 *lbs*

Total potassium generated: 127,583.99 *lbs*

Total salt generated: 720,848.28 *lbs*

	14,925,000	*gallons applied*
+	684,000	*gallons exported*
-	0	*gallons imported*
=	15,609,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
MID Lateral 3	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/07/2023	Corral solids	1,740.00 *ton*	Dry-weight	18.8		18,300.00	4,500.00	29,400.00		56.30
11/08/2023	Corral solids	540.00 *ton*	Dry-weight	18.8		18,300.00	4,500.00	29,400.00		56.30
11/10/2023	Corral solids	260.00 *ton*	Dry-weight	18.8		18,300.00	4,500.00	29,400.00		56.30

Date	Material type	Quantity	Kjeldahl-N (mg/L)	Ammonium-N (mg/L)	Ammonia-N (mg/L)	Nitrate-N (mg/L)	P (mg/L)	K (mg/L)	EC (µmhos/cm)	TDS (mg/L)
11/22/2023	Process wastewater sludge	684,000.00 *gal*	1,800.00	496.00	0.00	0.00	256.00	1,120.00		3,640

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	75,486.77	18,562.32	121,273.82	2,322,352.48
Process wastewater	10,274.36	1,461.24	6,392.94	20,777.05
Total exports for all materials	85,761.13	20,023.56	127,666.76	2,343,129.53

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Boonos	19	19	2	process wastewater	X012-X028-X021-XXXX
Dairy	12	12	2	process wastewater	X012-X028-X018-XXXX X012-X028-X042-XXXX X012-X028-X063-XXXX X012-X028-X064-XXXX
Edwin's	36	36	2	process wastewater	X012-X034-X005-XXXX X012-X034-X006-XXXX X012-X034-X007-XXXX X012-X034-X009-XXXX
Valley Oak	20	20	2	process wastewater	X012-X029-X026-XXXX X012-X029-X027-XXXX
Walther	43	43	2	process wastewater	X012-X028-X040-XXXX X012-X028-X041-XXXX
Totals for areas that were used for application	130	130	10		
Totals for areas that were not used for application					
Land application area totals	130	130	10		

B. CROPS AND HARVESTS

Boonos

Field name: Boonos

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 19 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	310.00 *ton*	As-is		65.0	6,730.00	1,160.00	8,440.00		12.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	16.32	219.61	37.85	275.41	1,416.21

Boonos

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 19 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	598.00 *ton*	As-is		71.6	3,490.00	520.00	5,110.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	100.80	0.00
Total actual harvest content	31.47	219.69	32.73	321.66	1,001.11

Dairy

Field name: Dairy

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 12 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	217.50 *ton*	As-is		73.6	4,110.00	860.00	6,580.00		10.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	18.13	148.99	31.18	238.53	1,033.56

06/05/2023: Corn, silage

Crop: Corn, silage Acres planted: 12 Plant date: 06/05/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	355.00 *ton*	As-is		69.0	3,380.00	480.00	5,200.00		7.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	29.58	199.98	28.40	307.67	1,283.92

Edwin's

Field name: Edwin's

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 36 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	742.00 *ton*	As-is		73.8	3,940.00	770.00	5,910.00		11.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	20.61	162.42	31.74	243.62	1,231.23

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 36 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	1,184.00 *ton*	As-is		70.4	3,290.00	570.00	4,230.00		4.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	32.89	216.41	37.49	278.24	798.28

Valley Oak

Field name: Valley Oak

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	366.00 *ton*	As-is		75.4	4,390.00	940.00	7,930.00		13.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	18.30	160.67	34.40	290.24	1,179.47

Valley Oak

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	606.00 *ton*	As-is		69.4	3,370.00	520.00	4,980.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	30.30	204.22	31.51	301.79	1,223.88

Walther

Field name: Walther

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 43 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	832.00 *ton*	As-is		74.6	5,640.00	900.00	6,920.00		12.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	19.35	218.25	34.83	267.79	1,199.16

06/05/2023: Corn, silage

Crop: Corn, silage Acres planted: 43 Plant date: 06/05/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	1,382.00 *ton*	As-is		69.8	3,650.00	550.00	5,750.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	32.14	234.62	35.35	369.60	1,261.80

NUTRIENT BUDGET

A. LAND APPLICATIONS

Boonos - 11/03/2022: Oats, silage-soft dough

Field name: Boonos

Crop: Oats, silage-soft dough — Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	72.18	11.38	152.19	724.70	330,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	0.00	2,805,000.00 *gal*
Application event totals		72.18	11.38	152.19	724.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	104.15	9.73	99.09	524.94	240,000.00 *gal*
Application event totals		104.15	9.73	99.09	524.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	70.49	6.94	114.24	745.78	300,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	41.44	2,550,000.00 *gal*
Application event totals		70.49	6.94	114.24	787.22	

Boonos - 06/03/2023: Corn, silage

Field name: Boonos

Crop: Corn, silage — Plant date: 06/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Boonos - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	91.64	9.03	148.51	969.51	390,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	53.87	3,315,000.00 *gal*
Application event totals		91.64	9.03	148.51	1,023.38	

07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	45.58	2,805,000.00 *gal*
Application event totals		0.00	0.00	0.00	45.58	

07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	66.01	6.54	135.72	785.31	300,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	41.44	2,550,000.00 *gal*
Application event totals		66.01	6.54	135.72	826.75	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	66.01	6.54	135.72	785.31	300,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	41.44	2,550,000.00 *gal*
Application event totals		66.01	6.54	135.72	826.75	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	41.44	2,550,000.00 *gal*
Application event totals		0.00	0.00	0.00	41.44	

Boonos - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	59.41	5.88	122.14	706.78	270,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	37.30	2,295,000.00 *gal*
Application event totals		59.41	5.88	122.14	744.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	35.22	2,167,500.00 *gal*
Application event totals		0.00	0.00	0.00	35.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	33.15	2,040,000.00 *gal*
Application event totals		0.00	0.00	0.00	33.15	

Dairy - 11/03/2022: Oats, silage-soft dough

Field name: Dairy

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	62.34	9.83	131.43	625.88	180,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	0.00	1,530,000.00 *gal*
Application event totals		62.34	9.83	131.43	625.88	

Dairy - 11/03/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	82.45	7.70	78.44	415.58	120,000.00 *gal*
Application event totals		82.45	7.70	78.44	415.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	55.81	5.50	90.44	590.41	150,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	32.81	1,275,000.00 *gal*
Application event totals		55.81	5.50	90.44	623.22	

Dairy - 06/05/2023: Corn, silage

Field name: Dairy

Crop: Corn, silage

Plant date: 06/05/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	94.87	9.35	153.75	1,003.69	255,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	55.77	2,167,500.00 *gal*
Application event totals		94.87	9.35	153.75	1,059.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	45.93	1,785,000.00 *gal*
Application event totals		0.00	0.00	0.00	45.93	

Dairy - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.94	6.73	139.67	808.21	195,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	42.65	1,657,500.00 *gal*
Application event totals		67.94	6.73	139.67	850.86	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.49	5.69	118.19	683.87	165,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	36.09	1,402,500.00 *gal*
Application event totals		57.49	5.69	118.19	719.96	

08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	39.37	1,530,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.37	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.26	5.17	107.44	621.70	150,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	32.81	1,275,000.00 *gal*
Application event totals		52.26	5.17	107.44	654.51	

09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	29.53	1,147,500.00 *gal*
Application event totals		0.00	0.00	0.00	29.53	

Dairy - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	26.25	1,020,000.00 *gal*
Application event totals		0.00	0.00	0.00	26.25	

Edwin's - 11/04/2022: Oats, silage-soft dough

Field name: Edwin's

Crop: Oats, silage-soft dough

Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	76.19	12.01	160.64	764.96	660,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	0.00	5,610,000.00 *gal*
Application event totals		76.19	12.01	160.64	764.96	

02/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.71	6.42	65.37	346.32	300,000.00 *gal*
Application event totals		68.71	6.42	65.37	346.32	

04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.41	7.33	120.59	787.21	600,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	43.74	5,100,000.00 *gal*
Application event totals		74.41	7.33	120.59	830.95	

Edwin's - 06/02/2023: Corn, silage

Edwin's - 06/02/2023: Corn, silage

Field name: Edwin's

Crop: Corn, silage

Plant date: 06/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	85.57	8.43	138.67	905.29	690,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	50.30	5,865,000.00 *gal*
Application event totals		85.57	8.43	138.67	955.60	

07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	47.02	5,482,500.00 *gal*
Application event totals		0.00	0.00	0.00	47.02	

07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	73.16	7.24	150.42	870.38	630,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	45.93	5,355,000.00 *gal*
Application event totals		73.16	7.24	150.42	916.31	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	69.68	6.90	143.26	828.94	600,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	43.74	5,100,000.00 *gal*
Application event totals		69.68	6.90	143.26	872.68	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	43.74	5,100,000.00 *gal*
Application event totals		0.00	0.00	0.00	43.74	

Edwin's - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	66.20	6.55	136.09	787.49	570,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	41.55	4,845,000.00 *gal*
Application event totals		66.20	6.55	136.09	829.04	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	41.55	4,845,000.00 *gal*
Application event totals		0.00	0.00	0.00	41.55	

09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	39.37	4,590,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.37	

Valley Oak - 11/04/2022: Oats, silage-soft dough

Field name: Valley Oak

Crop: Oats, silage-soft dough

Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.80	11.79	157.72	751.05	360,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	0.00	3,060,000.00 *gal*
Application event totals		74.80	11.79	157.72	751.05	

Valley Oak - 11/04/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.20	6.93	70.60	374.02	180,000.00 *gal*
Application event totals		74.20	6.93	70.60	374.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	66.97	6.60	108.53	708.49	300,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	39.37	2,550,000.00 *gal*
Application event totals		66.97	6.60	108.53	747.86	

Valley Oak - 06/02/2023: Corn, silage

Field name: Valley Oak

Crop: Corn, silage

Plant date: 06/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	80.36	7.92	130.23	850.19	360,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	47.24	3,060,000.00 *gal*
Application event totals		80.36	7.92	130.23	897.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	39.37	2,550,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.37	

Valley Oak - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	62.71	6.21	128.93	746.04	300,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	37.40	2,422,500.00 *gal*
Application event totals		62.71	6.21	128.93	783.44	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	59.58	5.90	122.48	708.74	285,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	37.40	2,422,500.00 *gal*
Application event totals		59.58	5.90	122.48	746.14	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	39.37	2,550,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.37	

08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	59.58	5.90	122.48	708.74	285,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	37.40	2,422,500.00 *gal*
Application event totals		59.58	5.90	122.48	746.14	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	35.43	2,295,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.43	

Valley Oak - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	31.49	2,040,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.49	

Walther - 11/03/2022: Oats, silage-soft dough

Field name: Walther

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	86.98	13.71	183.40	873.31	900,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	0.00	7,650,000.00 *gal*
Application event totals		86.98	13.71	183.40	873.31	

02/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	92.04	8.60	87.56	463.90	480,000.00 *gal*
Application event totals		92.04	8.60	87.56	463.90	

04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	87.21	8.59	141.34	922.69	840,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	51.27	7,140,000.00 *gal*
Application event totals		87.21	8.59	141.34	973.95	

Walther - 06/05/2023: Corn, silage

Walther - 06/05/2023: Corn, silage

Field name: Walther

Crop: Corn, silage

Plant date: 06/05/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	93.44	9.20	151.43	988.59	900,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	54.93	7,650,000.00 *gal*
Application event totals		93.44	9.20	151.43	1,043.52	

07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	51.27	7,140,000.00 *gal*
Application event totals		0.00	0.00	0.00	51.27	

07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	81.67	8.09	167.91	971.59	840,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	51.27	7,140,000.00 *gal*
Application event totals		81.67	8.09	167.91	1,022.86	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	75.84	7.51	155.92	902.19	780,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	47.61	6,630,000.00 *gal*
Application event totals		75.84	7.51	155.92	949.80	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	47.61	6,630,000.00 *gal*
Application event totals		0.00	0.00	0.00	47.61	

Walther - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	70.00	6.93	143.92	832.79	720,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	43.95	6,120,000.00 *gal*
Application event totals		70.00	6.93	143.92	876.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	43.95	6,120,000.00 *gal*
Application event totals		0.00	0.00	0.00	43.95	

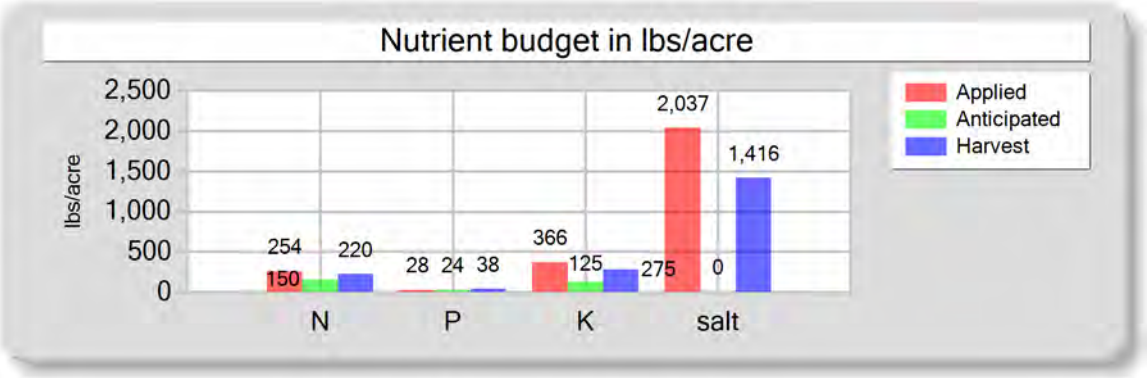
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	43.95	6,120,000.00 *gal*
Application event totals		0.00	0.00	0.00	43.95	

B. NUTRIENT BUDGET

Boonos - 11/03/2022: Oats, silage-soft dough

Field name: Boonos | Crop: Oats, silage-soft dough | Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	246.82	28.05	365.51	1,995.42
Fresh water	0.00	0.00	0.00	41.44
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	253.82	28.05	365.51	2,036.86
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	219.61	37.85	275.41	1,416.21
Nutrient balance	34.21	-9.80	90.10	620.65
Applied to removed ratio	1.16	0.74	1.33	1.44

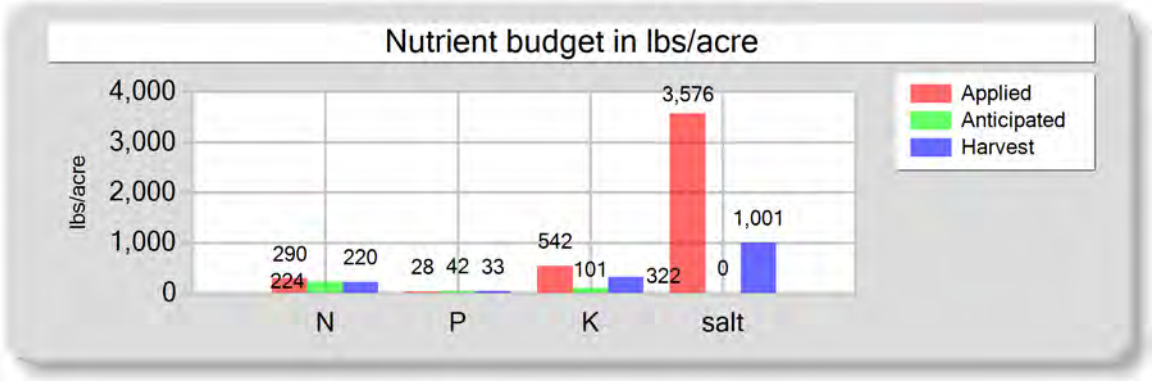
Fresh water applied	
5,355,000.00	*gallons*
197.21	*acre-inches*
10.38	*inches/acre*

Process wastewater applied	
870,000.00	*gallons*
32.04	*acre-inches*
1.69	*inches/acre*

Total harvests for the crop	
1	*harvests*

Boonos - 06/03/2023: Corn, silage

Field name: Boonos Crop: Corn, silage Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	283.08	27.98	542.09	3,246.91
Fresh water	0.00	0.00	0.00	329.44
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	290.08	27.98	542.09	3,576.35
Anticipated crop nutrient removal	224.00	42.00	100.80	0.00
Actual crop nutrient removal	219.69	32.73	321.66	1,001.11
Nutrient balance	70.39	-4.75	220.43	2,575.24
Applied to removed ratio	1.32	0.85	1.69	3.57

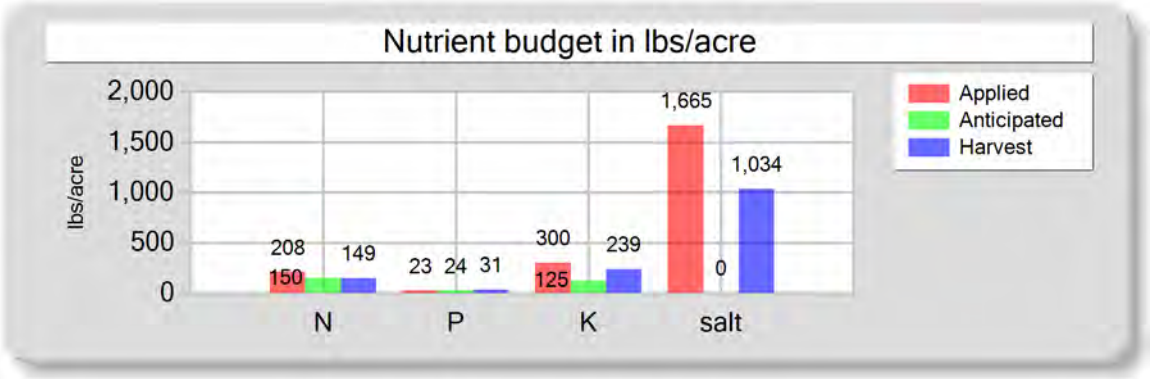
Fresh water applied	
20,272,500.00	*gallons*
746.57	*acre-inches*
39.29	*inches/acre*

Process wastewater applied	
1,260,000.00	*gallons*
46.40	*acre-inches*
2.44	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy - 11/03/2022: Oats, silage-soft dough

Field name: Dairy | Crop: Oats, silage-soft dough | Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	200.59	23.03	300.32	1,631.86
Fresh water	0.00	0.00	0.00	32.81
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	207.59	23.03	300.32	1,664.67
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	148.99	31.18	238.53	1,033.56
Nutrient balance	58.61	-8.15	61.79	631.11
Applied to removed ratio	1.39	0.74	1.26	1.61

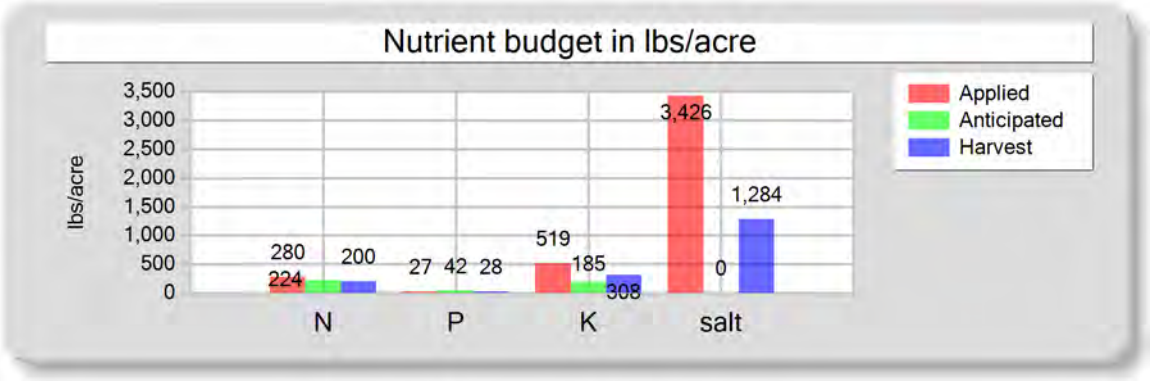
Fresh water applied	
2,805,000.00	*gallons*
103.30	*acre-inches*
8.61	*inches/acre*

Process wastewater applied	
450,000.00	*gallons*
16.57	*acre-inches*
1.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy - 06/05/2023: Corn, silage

Field name: Dairy | Crop: Corn, silage | Plant date: 06/05/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	272.56	26.94	519.05	3,117.48
Fresh water	0.00	0.00	0.00	308.38
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	279.56	26.94	519.05	3,425.86
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	199.98	28.40	307.67	1,283.92
Nutrient balance	79.57	-1.46	211.38	2,141.95
Applied to removed ratio	1.40	0.95	1.69	2.67

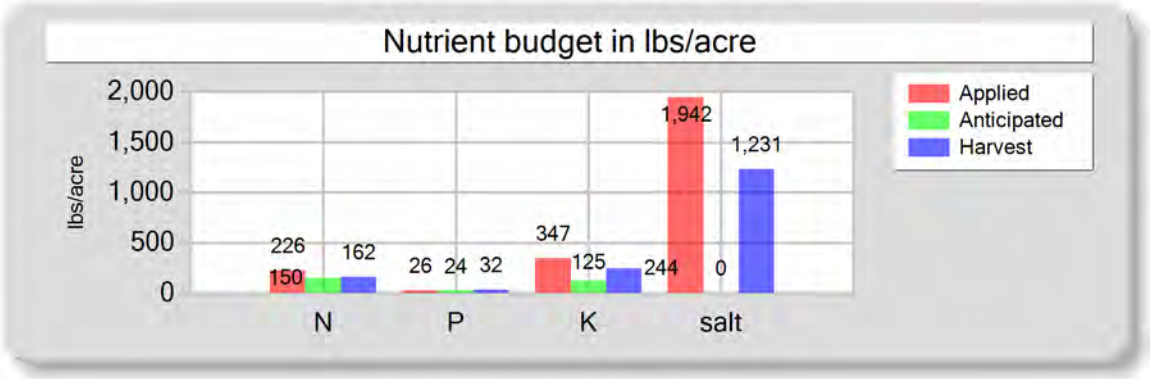
Fresh water applied	
11,985,000.00	*gallons*
441.37	*acre-inches*
36.78	*inches/acre*

Process wastewater applied	
765,000.00	*gallons*
28.17	*acre-inches*
2.35	*inches/acre*

Total harvests for the crop	
1	*harvests*

Edwin's - 11/04/2022: Oats, silage-soft dough

Field name: Edwin's Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	219.31	25.76	346.60	1,898.49
Fresh water	0.00	0.00	0.00	43.74
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	226.31	25.76	346.60	1,942.23
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	162.42	31.74	243.62	1,231.23
Nutrient balance	63.89	-5.98	102.97	711.00
Applied to removed ratio	1.39	0.81	1.42	1.58

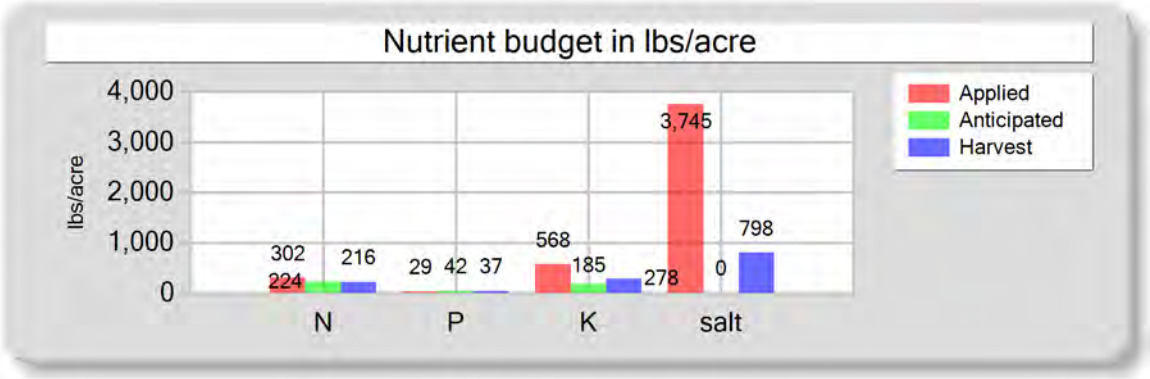
Fresh water applied	
10,710,000.00	*gallons*
394.41	*acre-inches*
10.96	*inches/acre*

Process wastewater applied	
1,560,000.00	*gallons*
57.45	*acre-inches*
1.60	*inches/acre*

Total harvests for the crop	
1	*harvests*

Edwin's - 06/02/2023: Corn, silage

Field name: Edwin's Crop: Corn, silage Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	294.61	29.12	568.44	3,392.10
Fresh water	0.00	0.00	0.00	353.21
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	301.61	29.12	568.44	3,745.32
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	216.41	37.49	278.24	798.28
Nutrient balance	85.20	-8.37	290.20	2,947.04
Applied to removed ratio	1.39	0.78	2.04	4.69

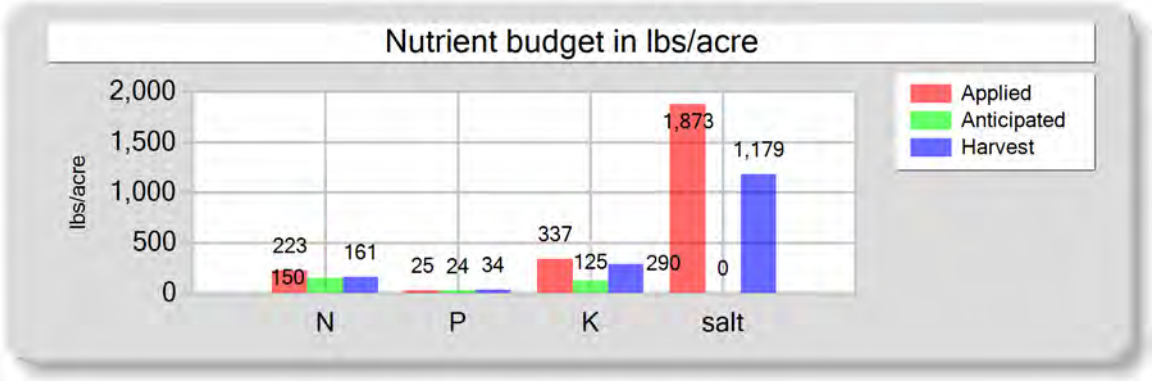
Fresh water applied	
41,182,500.00	*gallons*
1,516.61	*acre-inches*
42.13	*inches/acre*

Process wastewater applied	
2,490,000.00	*gallons*
91.70	*acre-inches*
2.55	*inches/acre*

Total harvests for the crop	
1	*harvests*

Valley Oak - 11/04/2022: Oats, silage-soft dough

Field name: Valley Oak | Crop: Oats, silage-soft dough | Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	215.98	25.32	336.85	1,833.56
Fresh water	0.00	0.00	0.00	39.37
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	222.98	25.32	336.85	1,872.93
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	160.67	34.40	290.24	1,179.47
Nutrient balance	62.30	-9.08	46.61	693.46
Applied to removed ratio	1.39	0.74	1.16	1.59

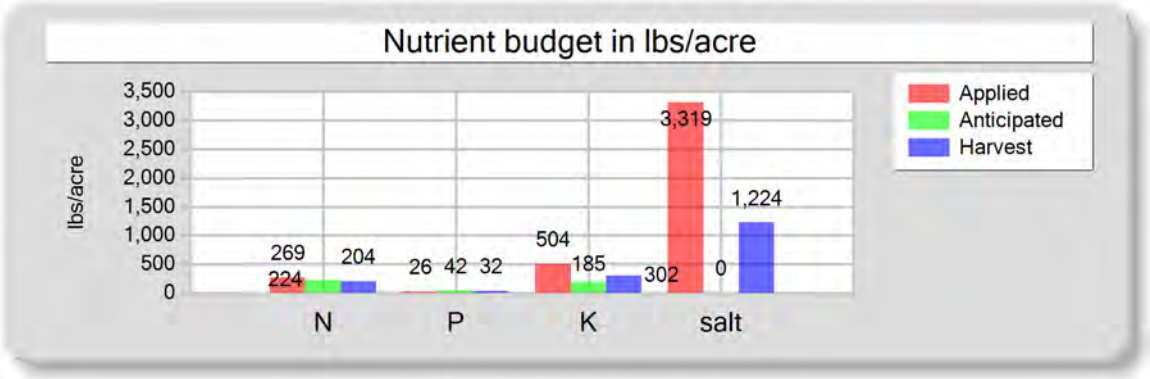
Fresh water applied	
5,610,000.00	*gallons*
206.60	*acre-inches*
10.33	*inches/acre*

Process wastewater applied	
840,000.00	*gallons*
30.93	*acre-inches*
1.55	*inches/acre*

Total harvests for the crop	
1	*harvests*

Valley Oak - 06/02/2023: Corn, silage

Field name: Valley Oak Crop: Corn, silage Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	262.23	25.92	504.13	3,013.71
Fresh water	0.00	0.00	0.00	305.10
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	269.23	25.92	504.13	3,318.81
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	204.22	31.51	301.79	1,223.88
Nutrient balance	65.01	-5.59	202.34	2,094.93
Applied to removed ratio	1.32	0.82	1.67	2.71

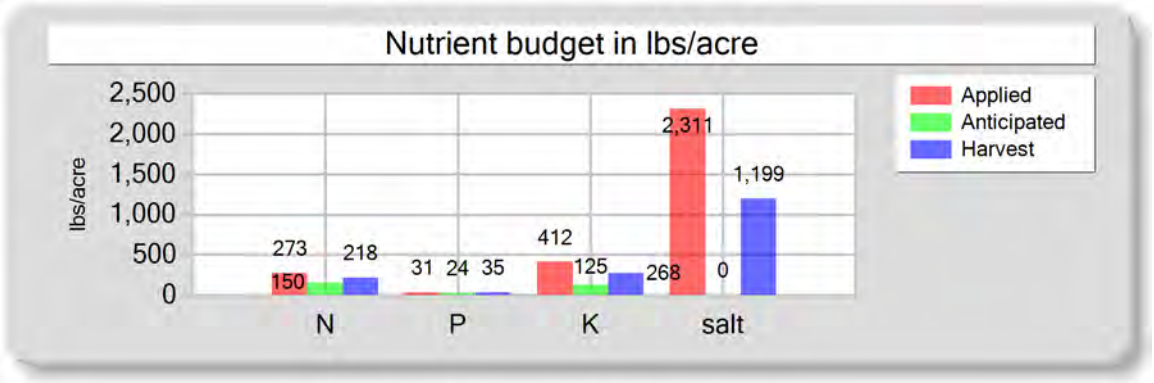
Fresh water applied	
19,762,500.00	*gallons*
727.79	*acre-inches*
36.39	*inches/acre*

Process wastewater applied	
1,230,000.00	*gallons*
45.30	*acre-inches*
2.26	*inches/acre*

Total harvests for the crop	
1	*harvests*

Walther - 11/03/2022: Oats, silage-soft dough

Field name: Walther | Crop: Oats, silage-soft dough | Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	266.23	30.90	412.30	2,259.90
Fresh water	0.00	0.00	0.00	51.27
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	273.23	30.90	412.30	2,311.17
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	218.25	34.83	267.79	1,199.16
Nutrient balance	54.98	-3.93	144.51	1,112.01
Applied to removed ratio	1.25	0.89	1.54	1.93

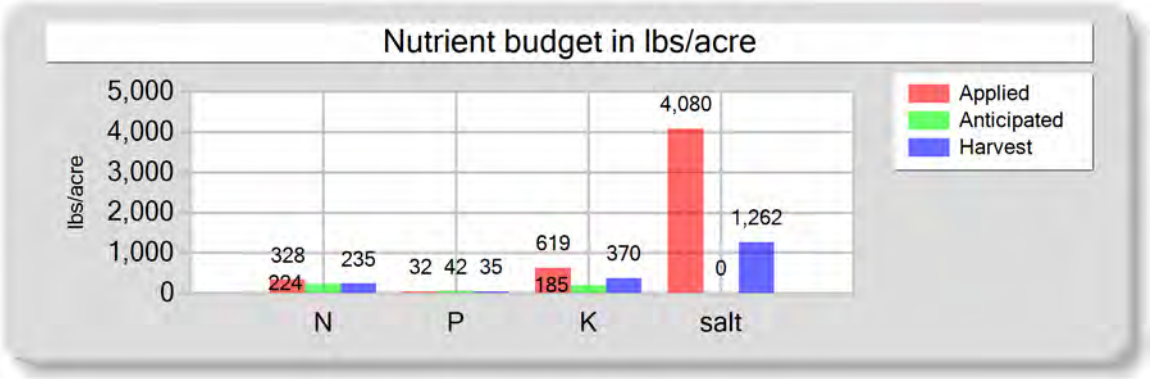
Fresh water applied	
14,790,000.00	*gallons*
544.67	*acre-inches*
12.67	*inches/acre*

Process wastewater applied	
2,220,000.00	*gallons*
81.76	*acre-inches*
1.90	*inches/acre*

Total harvests for the crop	
1	*harvests*

Walther - 06/05/2023: Corn, silage

Field name: Walther | Crop: Corn, silage | Plant date: 06/05/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	320.96	31.73	619.18	3,695.17
Fresh water	0.00	0.00	0.00	384.52
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	327.96	31.73	619.18	4,079.69
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	234.62	35.35	369.60	1,261.80
Nutrient balance	93.34	-3.62	249.57	2,817.89
Applied to removed ratio	1.40	0.90	1.68	3.23

Fresh water applied	
53,550,000.00	*gallons*
1,972.06	*acre-inches*
45.86	*inches/acre*

Process wastewater applied	
3,240,000.00	*gallons*
119.32	*acre-inches*
2.77	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Manure

Sample and source description: Corral Manure

Sample date: 05/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 30.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,200.00	5,020.00	15,400.00							75.60
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/13/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 18.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,300.00	4,500.00	29,400.00							56.30
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

WWQ4

Sample and source description: WWQ4

Sample date: 10/21/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	498.00	291.00	0.00	0.00	78.50	1,050.00								6,890.00	5,000
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Slurry

Sample and source description: Slurry

Sample date: 02/10/2023 Material type: Process wastewater sludge Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,800.00	496.00	0.00	0.00	256.00	1,120.00								6,330.00	3,640
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ1

Sample and source description: WWQ1

Sample date: 02/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	988.00	378.00	0.00	0.00	92.30	940.00								7,080.00	4,980
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 06/02/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	535.00	375.00	0.00	0.00	52.70	867.00	147.00	103.00	177.00	3,320.00	0.00	237.00	273.00	6,750.00	5,660
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 07/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	501.00	367.00	0.00	0.00	49.60	1,030.00								6,690.00	5,960
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 10/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	594.00	311.00	0.00	0.00	69.90	696.00								5,870.00	3,940
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

MID Lateral 3

MID Lateral 3

Sample description: MID Lateral 3

Sample date: 07/22/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	0
DL	0.50	0.50	0.50								1.00	10

MID Lateral 3

Sample description: MID Lateral 3

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								30.70	37
DL	1.00	0.50	0.50								10.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Boonos - 11/03/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,730.00	1,160.00	8,440.00		12.40
DL	100.00	100.00	100.00		0.01

Boonos - 06/03/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,490.00	520.00	5,110.00		5.60
DL	100.00	100.00	100.00		0.01

Dairy - 11/03/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,110.00	860.00	6,580.00		10.80
DL	100.00	100.00	100.00		0.01

Dairy - 06/05/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,380.00	480.00	5,200.00		7.00
DL	100.00	100.00	100.00		0.01

Edwin's - 11/04/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,940.00	770.00	5,910.00		11.40
DL	100.00	100.00	100.00		0.01

Edwin's - 06/02/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,290.00	570.00	4,230.00		4.10
DL	100.00	100.00	100.00		0.01

Valley Oak - 11/04/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 75.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,390.00	940.00	7,930.00		13.10
DL	100.00	100.00	100.00		0.01

Valley Oak - 06/02/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,370.00	520.00	4,980.00		6.60
DL	100.00	100.00	100.00		0.01

Walther - 11/03/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,640.00	900.00	6,920.00		12.20
DL	100.00	100.00	100.00		0.01

Walther - 06/05/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,650.00	550.00	5,750.00		6.50
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

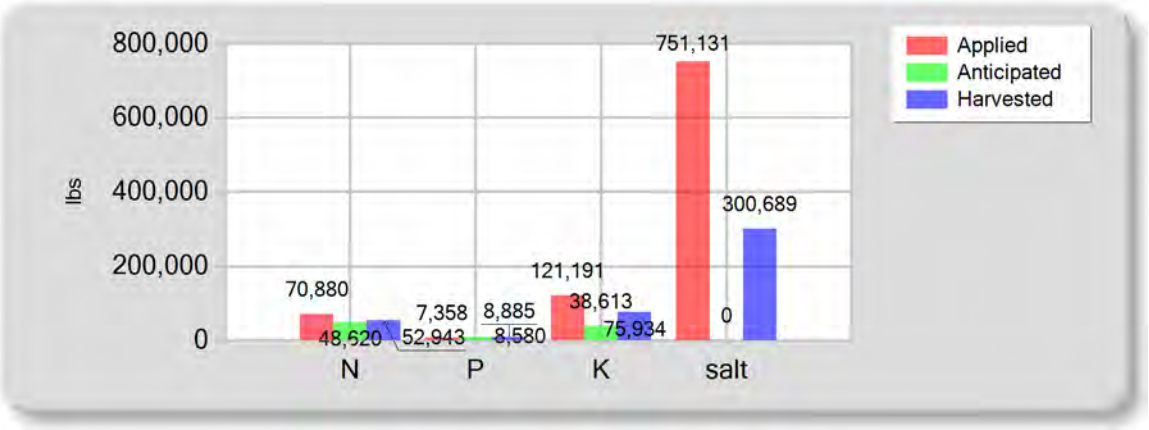
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

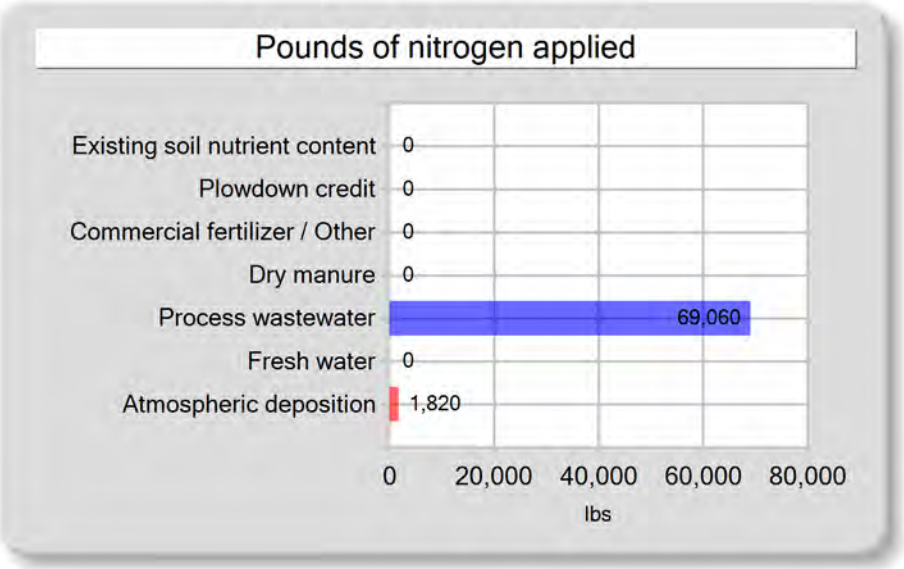
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

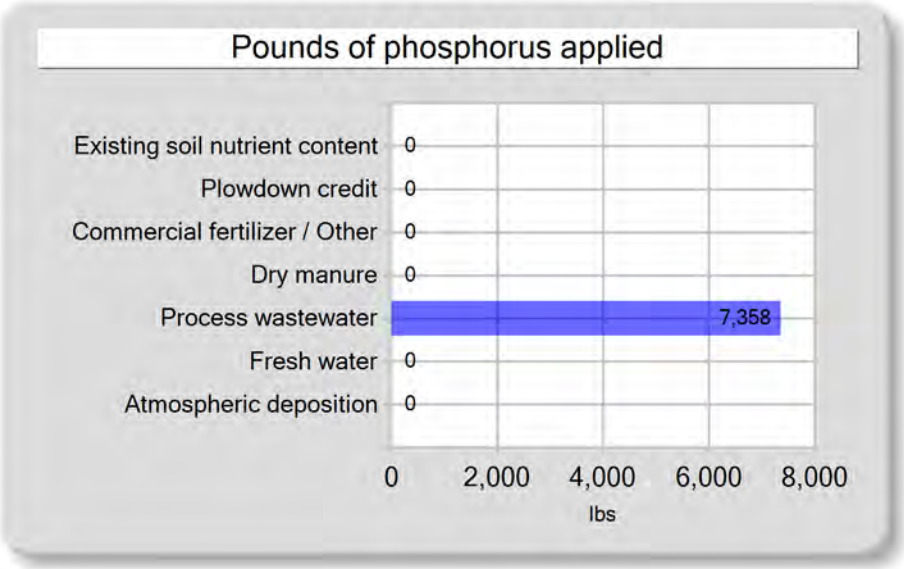
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	69,060.42	7,357.82	121,191.06	700,071.23
Fresh water	0.00	0.00	0.00	51,059.70
Atmospheric deposition	1,820.00	0.00	0.00	0.00
Total nutrients applied	70,880.42	7,357.82	121,191.06	751,130.93
Anticipated crop nutrient removal	48,620.00	8,580.00	38,613.00	0.00
Actual crop nutrient removal	52,943.45	8,884.58	75,934.14	300,689.40
Nutrient balance	17,936.97	-1,526.76	45,256.92	450,441.53
Applied to removed ratio	1.34	0.83	1.60	2.50

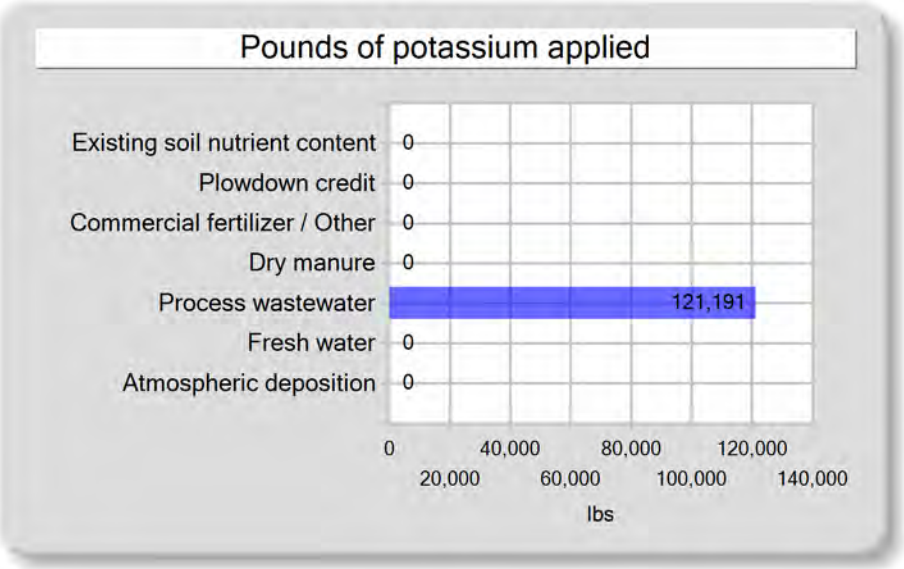
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

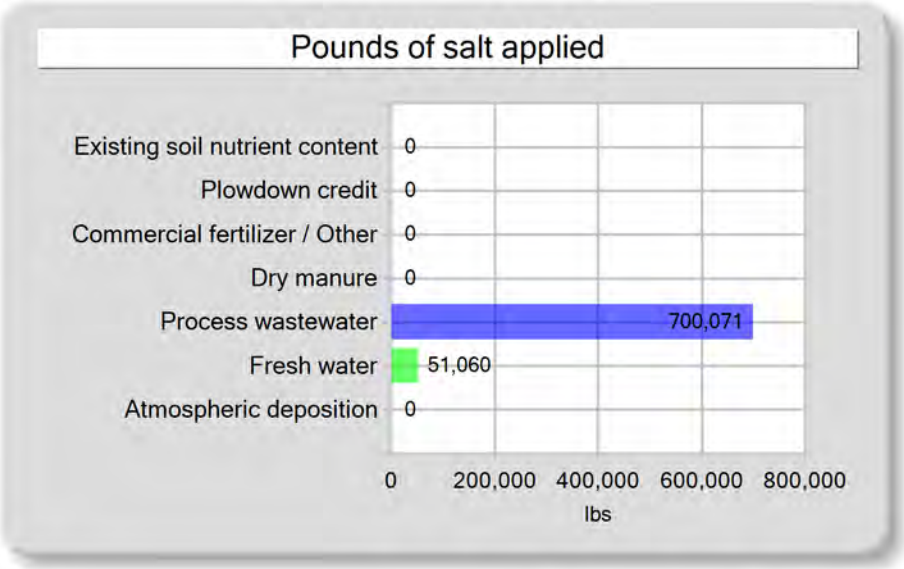


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Bill Genasci (Jun 24, 2024 14:01 PDT)	Frank Gwerder (Jun 21, 2024 21:16 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Bill Genasci	Frank Gwerder
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 24, 2024	Jun 21, 2024
DATE	DATE

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted.

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed.

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: FRANK GWERDER

Name of Dairy Facility: DOUBLE G DAIRY

Facility Address: 6555 BLUE GUM AVE. MODESTO 95358
Number and Street / City / Zip Code

Contact Person Name and Phone Number: FRANK GWERDER 209-579-5723
Name / Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: AVILA SPREADING SERVICE.

Address of Hauling Company /Person: P.O. BOX 580742 MODESTO 95358
Number and Street / City / Zip Code

Contact Person: LONNY AVILA 209-595-7955
Name / Phone Number

Destination Information:

Composting Facility / Broker / (Farmer) / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

IN + OUT LAND LEVELING 3642 FINNEY RD. MODESTO 95358 209-543-7546
Name / Number and Street / City / Zip Code / Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

3642 FINNEY RD. MODESTO 95358
Number and Street / City / Zip Code / Assessor's Parcel Number

Dates Hauled: 11/7/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 1740 (Tons) or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): 18.5% MOISTURE

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

174 LOADS @ 10 TONS PER LOAD.

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Frank Oberder (Jun 21, 2024 21:16 PDT) Date: Jun 21, 2024

Hauler's Signature: _____ Date: _____

AVILA
PROFESSIONAL · RELIABLE · COURTEOUS
SPREADING SERVICE
P.O. Box 580742 Modesto, California 95358

Frank Gwerder Dairy
1007 S Hart Rd
Modesto, Ca.95358

Date: 11-30-23
Terms: 10 Days
Fed ID: 77-0480211

Loads	Tons	Yards	Description	Cost
174	1740	3480	Manure taken by In N Out Land Leveling. Spread on Finney Rd fields	
54	540	1080	Manure taken by John Regusci	
26	260	520	Manure taken by Jared Aldrin, Patterson	

All manure taken from Gwerder Dairy, Blue Gum

254	**2540**	**5080**	**Total:**

Past due after 10 days from date. If account unpaid within time permitted, customer agrees to pay interest at 18% per annum and attorney's fees if account is referred to an attorney or collection agency for collection.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

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1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: FRANK GWERDER

Name of Dairy Facility: DOUBLE G DAIRY

Facility Address: 6555 BLUE GUM AVE. MODESTO 95358
Number and Street City Zip Code

Contact Person Name and Phone Number: FRANK GWERDER 209-579-5723
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: AVILA SPREADING SERVICE.

Address of Hauling Company /Person: P.O. BOX 580742 MODESTO 95358
Number and Street City Zip Code

Contact Person: LONNY AVILA 209-595-7955
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

JOHN REGUSCI 5425 STODDARD RD. MODESTO 95356 209-545-1277
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Number and Street City Zip Code 003-001-005 Assessor's Parcel Number

Dates Hauled: 11/8/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 540 Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): 18.8% MOISTURE
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

54 LOADS @ 10 TONS PER LOAD.

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Frank Carender (Jun 21, 2024 21:16 PDT) Date: Jun 21, 2024

Hauler's Signature: _____ Date: _____

AVILA
PROFESSIONAL · RELIABLE · COURTEOUS
SPREADING SERVICE

P.O. Box 580742 Modesto, California 95358

Frank Gwerder Dairy
1007 S Hart Rd
Modesto, Ca.95358

Date: 11-30-23
Terms: 10 Days
Fed ID: 77-0480211

Loads	Tons	Yards	Description	Cost
174	1740	3480	Manure taken by In N Out Land Leveling. Spread on Finney Rd fields	
54	540	1080	Manure taken by John Regusci	
26	260	520	Manure taken by Jared Aldrin, Patterson	

All manure taken from Gwerder Dairy, Blue Gum

254	**2540**	**5080**	**Total:**

Past due after 10 days from date. If account unpaid within time permitted, customer agrees to pay interest at 18% per annum and attorney's fees if account is referred to an attorney or collection agency for collection.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
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2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: FRANK GWERDER

Name of Dairy Facility: DOUBLE G DAIRY

Facility Address: 6555 BLUE GUM AVE. (Number and Street) MODESTO (City) 95358 (Zip Code)

Contact Person Name and Phone Number: FRANK GWERDER (Name) 209-579-5723 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: AVILA SPREADING SERVICE.

Address of Hauling Company /Person: P.O. BOX 580742 (Number and Street) MODESTO (City) 95358 (Zip Code)

Contact Person: LONNY AVILA (Name) 209-595-7955 (Phone Number)

Destination Information:

Composting Facility / (Broker) / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

LONNY AVILA (Name) P.O. BOX 580742 (Number and Street) MODESTO (City) 95358 (Zip Code) 209-595-7955 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

7243 PARADISE RD. (Number and Street) MODESTO (City) 95358 (Zip Code) _____ (Assessor's Parcel Number)

Dates Hauled: 11/10/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 260 (Tons) or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): 18.8% MOISTURE

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

26 LOADS @ 10 TONS PER LOAD.

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Frank Gilmender (Jun 21, 2024 21:16 PDT) Date: Jun 21, 2024

Hauler's Signature: _____ Date: _____

AVILA
PROFESSIONAL · RELIABLE · COURTEOUS
SPREADING SERVICE
P.O. Box 580742 Modesto, California 95358

Frank Gwerder Dairy
1007 S Hart Rd
Modesto, Ca.95358

Date: 11-30-23
Terms: 10 Days
Fed ID: 77-0480211

Loads	Tons	Yards	Description	Cost
174	1740	3480	Manure taken by In N Out Land Leveling. Spread on Finney Rd fields	
54	540	1080	Manure taken by John Regusci	
26	260	520	Manure taken by Jared Aldrin, Patterson	

All manure taken from Gwerder Dairy, Blue Gum

254	**2540**	**5080**	**Total:**

Past due after 10 days from date. If account unpaid within time permitted, customer agrees to pay interest at 18% per annum and attorney's fees if account is referred to an attorney or collection agency for collection.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: FRANK GWERDER

Name of Dairy Facility: DOUBLE G DAIRY

Facility Address: 6555 BLUE GUM AVE. (Number and Street) MODESTO (City) 95358 (Zip Code)

Contact Person Name and Phone Number: FRANK GWERDER (Name) 209-579-5723 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: FRANK GWERDER

Address of Hauling Company /Person: 825 S. HART RD. (Number and Street) MODESTO (City) 95358 (Zip Code)

Contact Person: FRANK GWERDER (Name) 209-579-5723 (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

MARCIA BOER (Name) 6413 SHOEMAKE AVE. (Number and Street) MODESTO (City) 95358 (Zip Code) 209-524-7266 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

_____ (Number and Street) _____ (City) _____ (Zip Code) 012-033-021 (Assessor's Parcel Number)

Dates Hauled: 11/22/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): _____

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 684,000 Gallons

Method used to determine volume of process wastewater: _____

72 LOADS OF SLURRY @ 9500 GALLONS PER LOAD.

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Frank Gravender (Jun 21, 2024 21:16 PDT) Date: Jun 21, 2024

Hauler's Signature: Frank Gravender (Jun 21, 2024 21:16 PDT) Date: Jun 21, 2024

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Double G Dairy- Modesto
1007 South Hart Rd
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 3/23/23 3:30 PM
SUBMITTED BY: Sean
ANALYZED DATE: 3/24/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 19.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW - Blue Gum	T3082015A	Electrical Conductivity (EC)	EPA 120.1 and	0.434	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	8.88	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	278	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	101			
Electrical Conductivity (EC)	102			3.27
Nitrate Nitrogen (NO3-N)	103	97.4 (T3082012A)	98.0 (T3082012A)	0.634

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Double G Dairy- Modesto
1007 South Hart Rd
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 3/23/23 3:30 PM
SUBMITTED BY: Sean
ANALYZED DATE: 3/24/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 19.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW - Milk Barn	T3082015B	Electrical Conductivity (EC)	EPA 120.1 and	0.432	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	8.89	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	276	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	101			
Electrical Conductivity (EC)	102			3.27
Nitrate Nitrogen (NO3-N)	103	97.4 (T3082012A)	98.0 (T3082012A)	0.634

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

865 S. KILROY ROAD
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: DOUBLE G DAIRY
Mailing address:
(not facility address) MODESTO CA
Phone#:

Authorized Copy Released To: F&R AG SERVICES
Sampled By: Sean Locke
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	3/23/23	12:15	DW-BLUE GUM	DOMESTIC	19.0°		X				
2	"	12:20	DW-MILK BARN	"	19.0°		X				
3											
4											
5											
6											
7											
8											
9											
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2023 Double G Dairy Annual Report

Final Audit Report 2024-06-24

Created:	2024-06-24
By:	Sean Locke (slocke@fragservices.com)
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